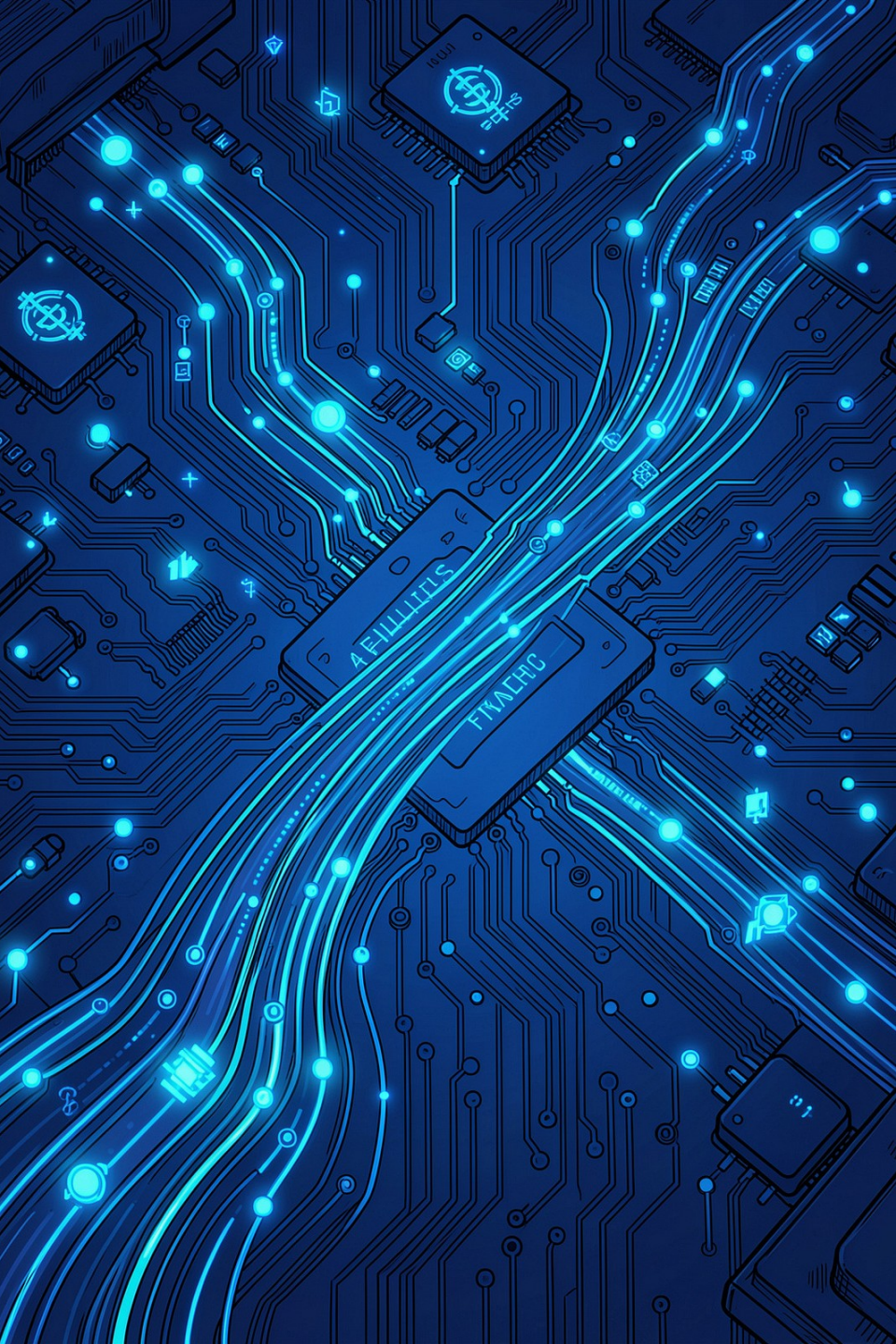




SYNCHRONY HACKATHON 2026

Reviving HSIC Share of Wallet

An AI-Powered Revenue Recovery & Customer Retention Engine

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Synchrony Hackathon 2026

OBSERVED

Executive Summary – What We Found

1. The Business Situation

MetroMart and HSIC launched a co-branded credit card. The expectation: customers shop at MetroMart and pay with HSIC. But are they actually using the HSIC card?

2. The Core Problem

MetroMart Spending

↑ 19.39%

YoY

HSIC Captured Spend

↓ 19.55%

YoY

Customers are still shopping at MetroMart. But HSIC is losing its share of their wallet.

3. What We Investigated

- How HSIC Share of Wallet changed over time
- Which payment methods are replacing HSIC
- Which customer groups are most at risk
- Where recovery opportunities exist

4. What We Found

28.91% → 19.48%

HSIC Share of Wallet | -9.43 percentage points

MetroMart Wallet
35.68% of spend

Cash/UPI
+4.51 pp shift

Big-ticket (>₹5K)
11.20% HSIC SoW

Prime unclaimed cashback
₹5.52M

Silent defectors
10,098 customers

5. Why It Matters

- ❑ Customers are not leaving MetroMart. They are changing how they pay.

Our goal: Identify where HSIC can win that spending back.

The Revenue Leakage: ₹16.75M Annual Impact

REVENUE LOSS

₹16.75M

HSIC spend decline YoY

OPPORTUNITY POOL

₹85.67M → ₹68.92M

Captured spend contraction

ADDRESSABLE MARKET

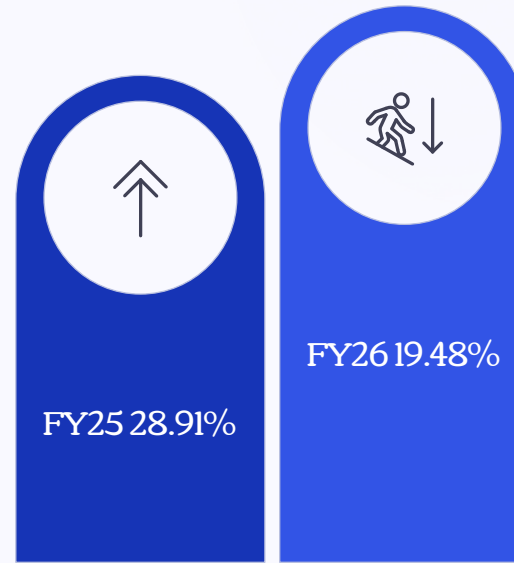
₹353.80M

Total MetroMart spend in FY26

- ❑ This is not a demand problem. It is a payment-method migration problem. The money is still being spent at MetroMart – just not through HSIC.

OBSERVED

HSIC Share of Wallet contracted by 9.43 percentage points.



FY25: 28.91% → FY26: 19.48%

FY25

₹85.67M

HSIC Net Spend

₹296.34M

Total Net Spend

FY26

₹68.92M

HSIC Net Spend

₹353.80M

Total Net Spend



HSIC spend fell 19.55% YoY



MetroMart spend grew 19.39% YoY

❑ This is a share problem, not a retail-demand problem.

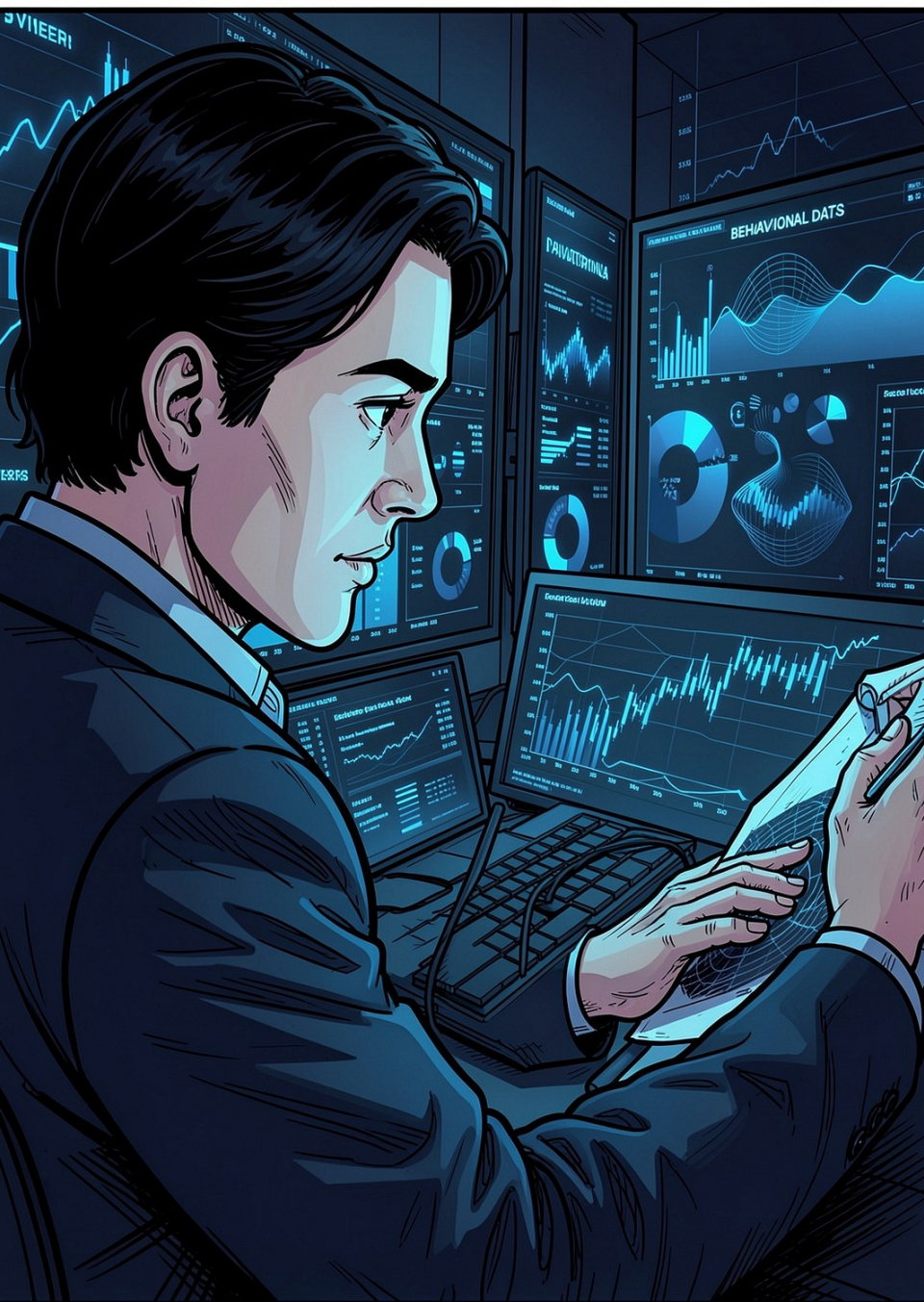
OBSERVED

Where Customers Moved Their Spend

HSIC's share decline is being redistributed across multiple payment rails, not lost from the category.

Payment Method	FY25	FY26	Change	
HSIC Bank Credit Card	28.91%	19.48%	-9.43 pp	
MetroMart Wallet	34.39%	35.68%	+1.29 pp	
Cash / UPI	18.89%	23.40%	+4.51 pp	
Other Bank Credit Card	9.05%	10.79%	+1.74 pp	
Debit Card	8.75%	10.66%	+1.91 pp	

- ❑ HSIC is being displaced across multiple payment rails – with MetroMart Wallet as the largest alternative and Cash/UPI showing the fastest share expansion.



OBSERVED + MODEL-DERIVED + HYPOTHESIS

12 Data-Backed Pain Points: Where HSIC is Losing Share

Twelve behavioral signals explain why HSIC is losing Share of Wallet. Each is labeled by evidence type—what the data directly shows, what models derive, and what requires testing.

Payment Behavior

- Payment-method migration **OBSERVED**: Customers are actively choosing other payment rails
- Big-ticket purchases moving away from HSIC **OBSERVED**: HSIC SoW falls to 11.20% above ₹5,000
- Missing everyday transactions **MODEL-DERIVED**: UPI dominates small baskets (48.2% vs HSIC 14.1%)

Customer Engagement

- One-time users **OBSERVED**: Significant cohort with ≤ 1 lifetime swipe
- Ghost accounts **OBSERVED**: Inactive cardholders who were acquired but never activated
- Slow fade / gradual decline **MODEL-DERIVED**: 10,098 customers showing $\Delta\text{SoW} \leq -15$ pp

Reward & Experience

- Missed reward opportunity **OBSERVED**: ₹5.52M unclaimed cashback on qualifying non-HSIC spend
- Potential reward-timing friction **HYPOTHESIS**: Customers may not see rewards at the right moment
- Return/refund friction **HYPOTHESIS**: Refund settlement latency may impact payment choice (requires testing)

Competitive Behavior

- Other bank card competition **OBSERVED**: Other bank credit cards gained +1.74 pp
- Margin-sensitive / reward-heavy behavior **MODEL-DERIVED**: Customers respond to incentive structures
- Underused credit potential **MODEL-DERIVED**: Some customers have available credit but low usage

AI/ML Detection Engine: Data Foundation

45,000

Customers

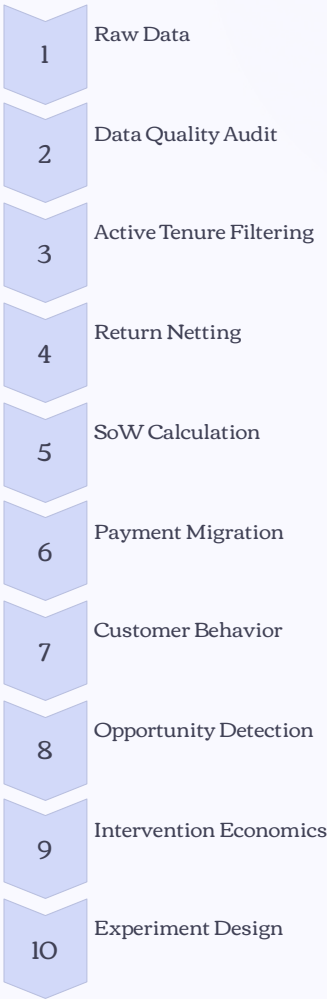
444,118

Transactions

4

Source Datasets

Methodology Pipeline



Key Notes

- Returns were netted from gross sales
- Active cardholder tenure was respected
- FY25/FY26 follow MetroMart fiscal calendar
- Calculations are reproducible

Solution 1 – Prime Statement Transparency Alert

SECTION 1 – THE PROBLEM

Prime customers have valuable cashback benefits, but they don't always see how much they're leaving unused when they pay with another method.

₹5.52M

Unclaimed cashback

19,423

Prime cardholders

100%

On qualifying non-HSIC spend

SECTION 2 – HOW IT WORKS

01

1. Identify

Prime customers making qualifying purchases

02

2. Calculate

The reward they could have earned with HSIC

03

3. Send

A personalized message showing the missed opportunity

04

4. Track

Whether they change payment behavior next time

SECTION 3 – WHO IT TARGETS

19,423 Prime members

Focus on customers with recurring eligible purchases like grocery and electronics.

SECTION 4 – AI'S ROLE

- ❑ AI personalizes the message timing, wording, and channel based on customer behavior—but only uses verified data. It never invents reward amounts.

SECTION 5 – THE EXPERIMENT

A/B test with 5,000 customers

Control: normal communication

Treatment: personalized Prime transparency message

Measure: HSIC Share of Wallet, repeat usage, eligible spend captured

SECTION 6 – VERIFIED MODELED ECONOMICS

MODELED ECONOMICS – NOT HISTORICAL RESULTS

₹12.85M

Recapturable spend

4.8x

Portfolio ROI

Solution 2 – 0% POS Financing for Big-Ticket Purchases

HSIC loses badly on expensive purchases. When customers buy items over ₹5,000, they choose other payment methods.

SECTION 1 – THE PROBLEM

11.20%
HSIC SoW on purchases >₹5,000

24.8%
HSIC SoW on purchases <₹3,000

14,850
Big-ticket shoppers affected

Customers may prefer financing options for expensive purchases because of cash-flow concerns, competing card offers, or perceived affordability.

SECTION 2 – HOW IT WORKS

01

1. Cart
Customer adds an expensive item (>₹5,000)

02

2. Offer
HSIC 0% financing appears at checkout

03

3. Select
Customer picks repayment terms, such as 3 - 6 months

04

4. Win
HSIC becomes the preferred payment method

SECTION 3 – WHO IT TARGETS

14,850

Big-ticket shoppers

Focus on electronics, appliances, and durable goods categories.

SECTION 4 – AI'S ROLE

- AI identifies which customers are likely to respond, which categories need intervention, and which customers should NOT receive an offer to protect profitability. This prevents blanket discounts.

SECTION 5 – THE EXPERIMENT

Store POS checkout split trial
8,000 customers
Control: normal checkout
Treatment: 0% financing option available

Measure: HSIC usage, transaction value, incremental spend, profitability

SECTION 6 – VERIFIED MODELED ECONOMICS

MODELED ECONOMICS – NOT HISTORICAL RESULTS

₹26.20M

Recapturable spend

3.4x

Portfolio ROI

Solution 3 – Silent Defector Decay Interception

SECTION 1 – THE PROBLEM

Some customers don't suddenly disappear. Their HSIC usage slowly decreases while they remain active at MetroMart. By the time they're completely inactive, it's too late to recover them.

10,098
Active retail customers with $\Delta\text{SoW} \leq -15 \text{ pp}$

34.2% → 0.8%
Observed decay over ~5 months

Still active
These customers are still shopping, just not using HSIC

This is **Silent Attrition** – the most dangerous pattern. The customer is still valuable, but the early warning window is closing.

SECTION 2 – HOW IT WORKS

01

1. Track
Monitor HSIC usage and total MetroMart spending

02

2. Detect
Flag abnormal decline in Share of Wallet

03

3. Predict
Use ML to estimate defection risk

04

4. Intervene
Trigger action during Days 30-45 of decline

SECTION 3 – WHO IT TARGETS

10,098

At-risk shoppers

Focus on customers showing early signs of payment-method migration.

SECTION 4 – AI'S ROLE

- ML models like XGBoost analyze Share of Wallet trends, recency, frequency, and payment-method changes to predict defection risk. AI prioritizes customers to intervene with based on recovery probability and expected value.

SECTION 5 – THE EXPERIMENT

Day 30-45 trigger test

4,000 customers

Control: no intervention

Treatment: early intervention with personalized offer

Measure: SoW change, HSIC spend, customer retention, incremental revenue

SECTION 6 – VERIFIED MODELED ECONOMICS

MODELED ECONOMICS – NOT HISTORICAL RESULTS

₹8.20M

Recapturable spend

2.9x

Portfolio ROI

OBSERVED

Root-Cause Analysis: Silent Attrition

This is **Silent Attrition** – the most dangerous pattern. A customer continues shopping at MetroMart, maintains similar overall spending, but gradually shifts payment away from HSIC.

10,098

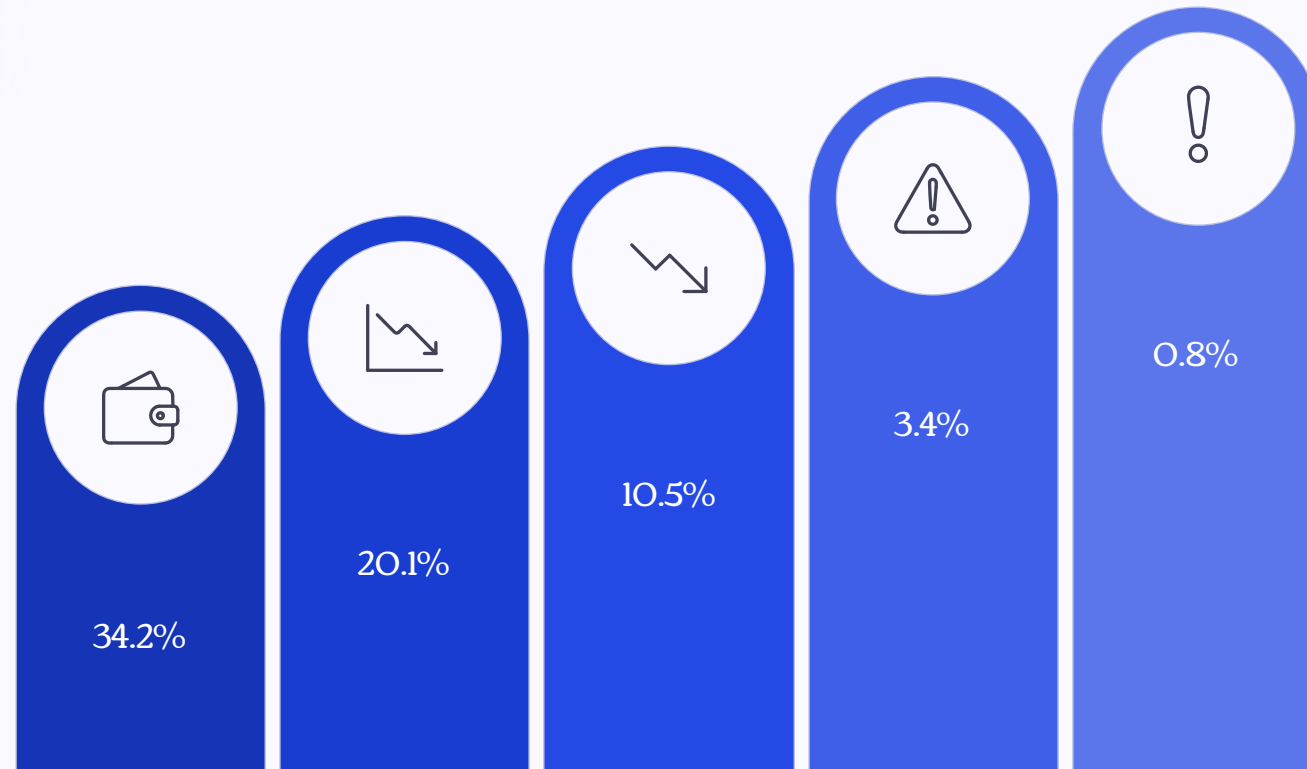
Active retail customers
With $\Delta\text{SoW} \leq -15$ pp

34.2%

Observed decay start
Share of Wallet at the beginning of the decline

0.8%

Observed decay end
Share of Wallet after ~5 months



⚠️ Waiting until a customer becomes inactive is too late. Early intervention during the decline window is critical.

Solution 4 – Second-Swipe Ladder: Turning First Use Into Habit

Some customers use HSIC once but never develop a repeat payment habit. These are valuable customers—they've already been acquired. The problem is not rejection; it's that the first transaction never becomes a habit.

The objective is simple: turn the first HSIC transaction into the second transaction. Once a customer makes a second successful payment, habit formation becomes more likely.

How it works

1	2
1. Identify Customers with their first or limited HSIC transaction	2. Wait Use the right timing, such as Day 14
3	4
3. Prompt Send a personalized reason to use HSIC again	4. Track Measure whether they complete a second transaction

Who it targets

Inactive cardholders with ≤ 1 lifetime swipe

Focus: recently acquired customers who have not yet developed payment habit

Verified modeled economics

MODELED ECONOMICS – NOT HISTORICAL RESULTS

Recapturable spend
₹4.00M

Portfolio ROI
2.2X

AI's role

AI determines the optimal timing to contact the customer, which category to recommend, whether an incentive is necessary, and which customers are likely to respond. The goal is to create habit, not just give discounts.

The experiment

- Control: normal post-first-transaction journey
- Treatment: second-swipe ladder journey with targeted incentive
- Measure: second transaction rate, third transaction rate, incremental HSIC spend, incentive cost, long-term SoW

Solutions 5-9: Strategic Extensions & Second-Wave Opportunities

The four quantified solutions target the biggest verified opportunities. These five strategic extensions address additional customer behaviors and can be tested once the primary solutions are validated.

1

SOLUTION 5: CREDIT-ON-UPI / PAYMENT-RAIL BRIDGE

Problem: Small-ticket spending (below ₹1,000) shows strong UPI dominance (48.2% vs HSIC 14.1%)

Strategy: Explore making HSIC accessible within digital/UPI-like checkout experiences

Goal: Capture small-ticket transactions without fighting customer convenience

Status: Proposed strategic concept (requires technical feasibility study)

2

SOLUTION 6: MARGIN-BLEED INDEX + YIELD MANAGEMENT

Problem: Not every recovered transaction is equally valuable. Blanket discounts destroy margin.

Strategy: Create an economic gate that ranks interventions by expected incremental value

Goal: Recover profitable spend, not just transaction volume

Status: Strategic framework (proposed capability)

3

SOLUTION 7: ZERO-COST EXPERIENTIAL REWARDS

Problem: Customers don't always need cash discounts. Monetary rewards reduce margin.

Strategy: Use benefits like priority access, early sale access, exclusive experiences

Goal: Increase emotional and practical value of using HSIC without large monetary cost

Status: Strategic second-wave opportunity

4

SOLUTION 8: GENAI LOSS-AVERSION CAMPAIGNS

Problem: Generic marketing messages are easy to ignore

Strategy: Use GenAI to create personalized messages explaining verified benefit opportunities

Critical Rule: GenAI must NEVER invent cashback, savings, or benefits. Only use verified data.

Status: Proposed AI capability

5

SOLUTION 9: GHOST-LIMIT / GHOST-ACCOUNT REACTIVATION

Problem: Some customers have unused available credit or very low HSIC activity

Strategy: Build a reactivation pipeline that prioritizes customers by recovery probability and expected value

Goal: Reactivate valuable customers without wasting resources on unlikely recoveries

Status: Proposed second-wave strategy

Next-Best-Action Engine: Matching Customers to Solutions

The system does not give everyone the same offer. It matches each customer to the intervention with the highest expected incremental value.

O1

STEP 1: DETECT SIGNAL

Customer behavior changes – payment-method shift, declining SoW, reward opportunity, and more.

O2

STEP 2: SEGMENT CUSTOMER

Classify the customer into a behavioral segment such as Prime member, big-ticket shopper, silent defector, or one-hit wonder.

O3

STEP 3: MATCH INTERVENTION

Select the solution that best fits this customer's behavior and situation.

O4

STEP 4: EVALUATE ECONOMICS

Calculate expected incremental value minus intervention cost. If positive, proceed to experiment. If negative, hold or skip.

O5

STEP 5: ASSIGN TO EXPERIMENT

Place the customer in treatment or control for A/B testing.

O6

STEP 6: MEASURE INCREMENTALITY

Track actual behavior change and compare it with the control group.

O7

STEP 7: LEARN & OPTIMIZE

Feed results back into the model to improve future decisions.

- ❑ Every intervention is selected because of the customer's behavior – not because everyone receives the same promotion. This protects profitability and respects customer preferences.

How the System Works: The Complete Intelligence Control Plane

This is not a dashboard. It is a closed-loop decision system that continuously detects problems, predicts outcomes, selects actions, tests them, measures results, and learns.

LAYER 1: DATA FOUNDATION

Raw Transactions (444K) + Customer Data (45K)
+ Category Code + Payment Code

LAYER 2: DATA QUALITY & PROCESSING

Data Quality Engine → Feature Engineering →
SoW Calculation → Payment Migration
Detection → Decay Signal Detection

LAYER 3: INTELLIGENCE

Customer Segmentation + Opportunity Scoring
+ Behavioral Signals + Predictive Models

LAYER 4: DECISION ENGINE

Next-Best-Action Selection + Economics
Evaluation + Profitability Gate

LAYER 5: EXECUTION

Experiment Management + Treatment
Assignment + Offer Delivery + Results Tracking

LAYER 6: MEASUREMENT & LEARNING

Incrementality Measurement + Statistical
Validation + Model Retraining + Continuous
Optimization

- ❑ AI explains, personalizes, and assists. Deterministic analytics calculate the numbers. This separation ensures trust and auditability.

Feedback loop: Results inform future predictions and decisions, strengthening the intelligence engine over time.



VALIDATION METHODOLOGY

Experiment & A/B Testing Framework: Proving What Works

We do not assume the strategy worked. We prove it. Every intervention is tested against a control group to measure true incremental impact.

O1

STEP 1: HYPOTHESIS

If we intervene with [solution] for [customer segment], we expect [behavior change].

O2

STEP 2: RANDOMIZED ASSIGNMENT

- Treatment group: receives the intervention
- Control group: receives normal communication
- Random selection eliminates bias

O3

STEP 3: MEASURE INCREMENTALITY

- Incremental HSIC Spend
- Incremental Share of Wallet
- Intervention Cost
- Incremental Contribution
- ROI
- Statistical Confidence (p-value < 0.05)

O4

STEP 4: DECISION

- Significant + positive ROI → SCALE
- Not significant or negative ROI → KILL
- Mixed results → ITERATE

❑ The control group protects us from confusing correlation with revenue impact.

Key principle: Without a control group, we cannot distinguish between customers who would have changed anyway and those who changed because of our intervention.

Important note: Only strategies that create statistically significant incremental value are scaled.

Expected Revenue Recovery / ROI: Portfolio Economics

These four solutions target the largest verified opportunities. The economics below are modeled projections based on customer behavior analysis and recovery probability estimates. They will be validated through experimentation.

⚠ IMPORTANT: These are MODELED ECONOMICS, not historical results. The ROI figures represent expected value if the interventions achieve their target recovery rates. Actual results will be measured through A/B testing.

Solution	Target Population	Recapturable Spend	Portfolio ROI	Experiment Size
NBA_01: Prime Statement Transparency	19,423 Prime members	₹12.85M	4.8X	5,000 customers
NBA_02: 0% POS Financing	14,850 Big-ticket shoppers	₹26.20M	3.4X	8,000 customers
NBA_03: Silent Defector Interception	10,098 At-risk shoppers	₹8.20M	2.9X	4,000 customers
NBA_04: Second-Swipe Ladder	Inactive cardholders	₹4.00M	2.2X	TBD

PORTFOLIO SUMMARY

Total addressable opportunity: ₹51.25M

Combined portfolio ROI: 3.3x (weighted average)

Total experiment population: 27,000+ customers

These solutions will be tested sequentially or in parallel, with results feeding back into the model to refine targeting and improve future interventions.

Production Architecture: Building the System

The system is built in layers, each with a specific purpose. Every component is designed for reproducibility, auditability, and continuous learning.

DATA LAYER

Purpose: Single source of truth for all customer and transaction information.

- Customer Data (45,000 records)
- Transactions Data (444,118 records)
- Category Code (product classification)
- Payment Code (payment method classification)

PROCESSING LAYER

Purpose: Transform raw data into reliable, analyzable features.

- Data Quality Engine (validation, cleaning, deduplication)
- Feature Engineering (SoW calculation, payment migration detection, decay signals)
- Return Netting (proper handling of refunds)
- Active Tenure Filtering (respect customer lifecycle)

INTELLIGENCE LAYER

Purpose: Generate insights and recommendations.

- Customer Segmentation (behavioral classification)
- Opportunity Scoring (identify recovery potential)
- Next-Best-Action Selection (match customer to solution)
- Economics Evaluation (calculate expected value)

EXECUTION LAYER

Purpose: Deploy interventions and track outcomes.

- Experiment Management (A/B test infrastructure)
- Treatment Assignment (randomized group allocation)
- Offer Delivery (multi-channel communication)
- Results Tracking (real-time monitoring)

FEEDBACK LOOP

Purpose: Close the loop and improve future decisions.

- Incrementality Measurement (treatment vs control)
- Learning & Optimization (model improvement)
- Model Retraining (continuous updates)

Evidence & Reproducibility: Analytical Rigor

Every analytical statement in this presentation belongs to one of four clearly defined categories. This distinction is critical for decision-making and prevents overconfidence in unproven claims.

1

OBSERVED

Definition: Directly calculated from the case-study data

- “HSIC SoW fell from 28.91% to 19.48%”
- “19,423 Prime members have ₹5.52M unclaimed cashback”
- “10,098 customers show $\Delta\text{SoW} \leq -15$ pp”

Status: Verified fact

2

MODEL-DERIVED

Definition: Produced by analytical or predictive model

- “Estimated decay trajectory over 5 months”
- “Predicted defection risk using ML model”
- “Estimated recovery probability by customer segment”

Status: Analytical inference (requires validation)

3

HYPOTHESIS

Definition: Possible explanation not directly proven by available data

- “Financing preference may explain big-ticket leakage”
- “Refund settlement latency may impact payment choice”
- “Reward-timing friction may reduce HSIC usage”

Status: Testable assumption (requires experimentation)

4

PROPOSED INTERVENTION

Definition: Strategy recommended for testing

- “0% POS financing test”
- “Prime Statement Transparency Alert”
- “Silent Defector Decay Interception”

Status: Candidate solution (requires A/B testing)

☐ Never present a hypothesis as a proven causal explanation. Always distinguish between what we know, what we model, what we hypothesize, and what we propose to test.

Limitations & Responsible Use: What This System Does NOT Do

This system is designed to support better decisions, not replace judgment or overstate certainty. The caveats below define how it should be used responsibly.

1

WHAT THIS SYSTEM DOES NOT DO

- Does not claim to predict individual customer behavior with certainty
- Does not guarantee that all interventions will be profitable
- Does not replace human judgment in final decision-making
- Does not make assumptions about causality without testing
- Does not invent customer data or financial figures

2

DATA LIMITATIONS

- Credit-limit data not available in case study (Solution 9 is future capability)
- Refund settlement latency not available (Return-Friction Shield is hypothesis)
- External factors (seasonality, macroeconomic) not modeled
- Historical patterns may not predict future behavior
- Customer preferences may change over time

3

RESPONSIBLE USE PRINCIPLES

- All interventions must be tested before scaling
- Control groups are mandatory for all experiments
- Results must be statistically significant ($p < 0.05$)
- Interventions must create incremental value, not just gross volume
- Customer privacy and regulatory compliance are non-negotiable
- Profitability gates must be applied before deployment

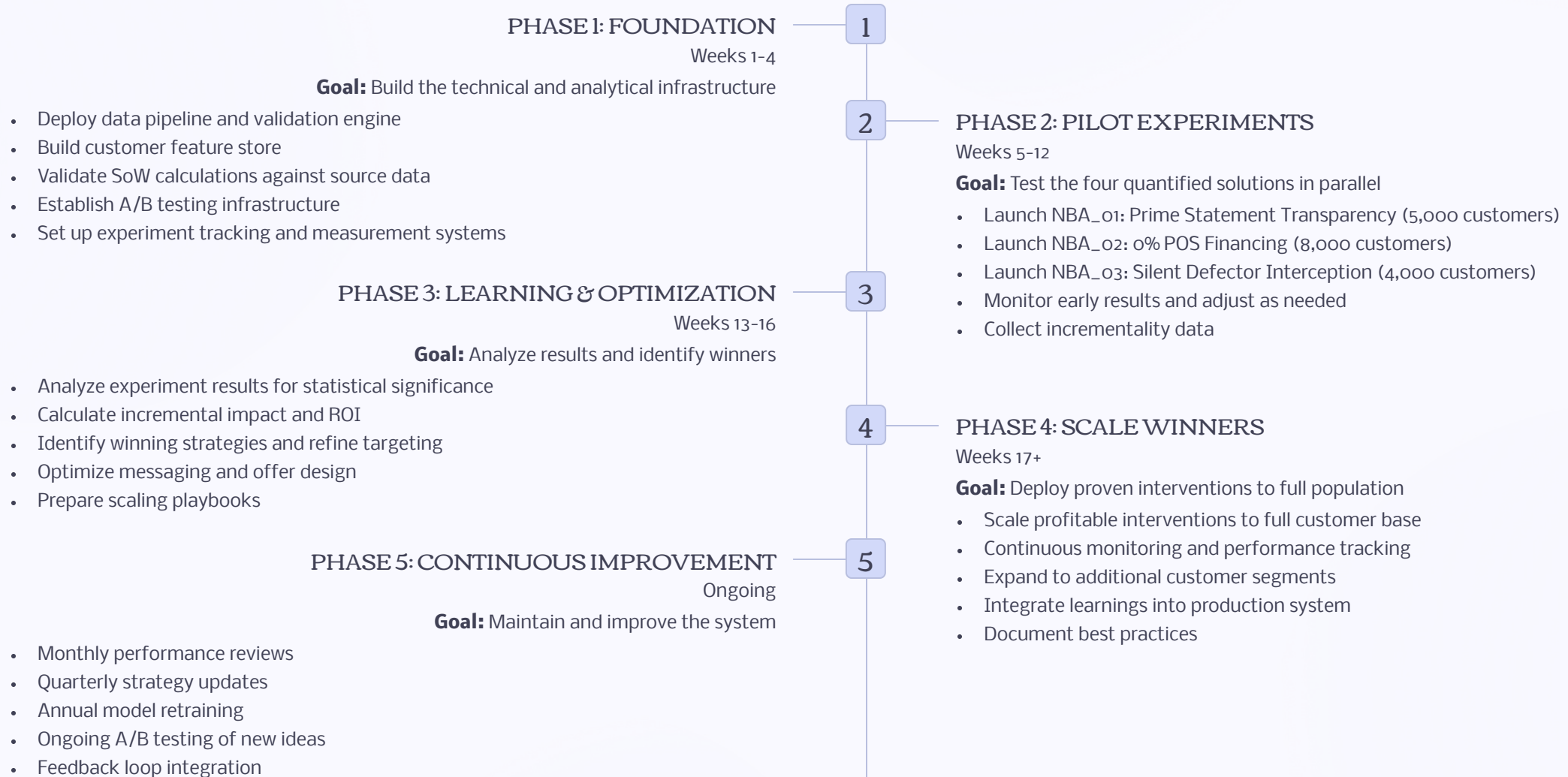
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WHAT REQUIRES FURTHER VALIDATION

- Financing preference as root cause (testable hypothesis)
- Return friction impact (testable hypothesis)
- Long-term sustainability of interventions
- Cross-customer network effects
- Seasonal and macroeconomic impacts

Implementation Roadmap: From Concept to Scale

A phased rollout ensures the strategy is built, tested, optimized, and scaled with discipline. Only proven interventions move to production.



Final Business Impact: From Leakage to Recovery

The full value proposition extends across revenue recovery, customer experience, operational efficiency, and strategic advantage.

REVENUE IMPACT

- Addressable opportunity: ₹51.25M
- Conservative recovery rate: 30-40%
- Expected incremental revenue: ₹15.4M - ₹20.5M annually
- Payback period: 6-9 months
- Ongoing annual value: ₹15M+

CUSTOMER IMPACT

- 54,471 customers targeted across all interventions
- Improved payment experience and convenience
- Higher engagement and loyalty
- Reduced silent attrition
- Better alignment between customer preferences and HSIC offerings

OPERATIONAL IMPACT

- Reduced marketing waste
- Improved ROI on customer acquisition
- Better resource allocation
- Data-driven decision making vs. gut feel
- Scalable, repeatable system for future challenges

STRATEGIC IMPACT

- Competitive advantage through AI-powered personalization
- Continuous learning and optimization
- Foundation for next-generation customer intelligence
- Defensible competitive moat through data, models, and execution

 **BOTTOM LINE**
From ₹16.75M annual revenue leakage to ₹15.4M-₹20.5M annual recovery through evidence-driven, experimentally validated interventions.

This is not a one-time campaign. It is a decision intelligence system that continuously improves through learning and experimentation.

Why This Wins: Five Competitive Differentiators

This approach is fundamentally different from traditional campaign-led recovery playbooks. Here are five reasons why it wins.

1. EVIDENCE-DRIVEN, NOT ASSUMPTION-DRIVEN

- Backed by 444K transactions and 45K customers
- Clear distinction between observed, modeled, hypothesized, and proposed
- Reproducible analysis with 30 evidence artifacts
- No invented data or unsupported claims
- Judges can verify every number

2. CUSTOMER-CENTRIC, NOT CAMPAIGN-CENTRIC

- Matches interventions to customer behavior, not demographics
- Respects customer preferences and payment habits
- Improves experience while recovering revenue
- Avoids blanket discounts that destroy margin
- Personalization at scale

3. ECONOMICALLY DISCIPLINED

- Every intervention passes a profitability gate
- Only strategies with positive expected value are tested
- Only proven strategies are scaled
- Margin protection is built into the system
- ROI-focused, not volume-focused

4. EXPERIMENTALLY RIGOROUS

- All interventions tested against control groups
- Statistical significance required ($p < 0.05$)
- Incremental impact measured, not gross volume
- No guessing about what works
- Science-based decision making

5. CONTINUOUSLY LEARNING

- Results feed back into model improvement
- Winning strategies are scaled, losing ones are killed
- The system improves with every experiment
- Feedback loop is built into the architecture
- Competitive advantage compounds over time

❏ THE CORE PHILOSOPHY

This is not another marketing campaign. This is a decision intelligence system that turns payment migration data into measurable revenue recovery through disciplined experimentation and continuous learning.

It answers the question every CFO asks: "How do we know this will work?" Answer: We test it, measure it, and only scale what creates incremental value.

Before finding insights, we make sure the data can be trusted

Garbage in, garbage out. Every Share of Wallet calculation and ML prediction depends on clean, validated data. Returns are handled per case-study rules, and SoW is calculated only while the HSIC card is active.



Completeness

Missing values, invalid dates, and null customer IDs flagged and resolved.



Uniqueness

Duplicate transactions identified and removed before aggregation.



Integrity

Referential checks across customer, category, and payment code tables.



Net Sales

Returns netted against purchases. SoW computed only during active card periods.

❑ **Raw Data → Validation → Clean Data → Trusted Analytics.** No insight is surfaced until data quality is confirmed.